

Safety data sheet

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Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)

Date / Revised: 01.09.2025

Version: 0.0

Product: **Tonalin® FFA 80**

(ID no. 30531864/SDS_GEN_00/EN)

Date of print 01.09.2025

1. Identification

Product identifier

Tonalin® FFA 80

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Raw material for the food industry

Details of the supplier of the safety data sheet

Company: Louis Dreyfus Company Ingredients GmbH
Robert-Hansen-Straße 1, 89257 Illertissen, Germany
+4973031664697

Emergency telephone number

International emergency number:
Telephone Germany toll free 0800 1817059. International Chemtrec number: +44 20 3885 0382

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Aquatic Chronic 3
Skin Irrit. 3

For the classifications not written out in full in this section the full text can be found in section 16.

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Label elements

Globally Harmonized System (GHS)

Signal Word:

Warning

Hazard Statement:

H316

Causes mild skin irritation.

H412

Harmful to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P273

Avoid release to the environment.

Precautionary Statements (Response):

P332 + P313

If skin irritation occurs: Get medical attention.

Precautionary Statements (Disposal):

P501

Dispose of contents and container to hazardous or special waste collection point.

Other hazards

According to UN GHS criteria

When finely distributed on porose material, self-ignition is possible. High risk of slipping due to leakage/spillage of product.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

Preparation based on: 10,12-Octadecadienoic acid, (10E,12Z)-, 9,11-Octadecadienoic acid, (9Z,11E)-

Hazardous ingredients (GHS)

According to UN GHS criteria

10,12-Octadecadienoic acid, (10E,12Z)-

Content (W/W): $\geq 25\%$ - $< 50\%$ CAS Number: 2420-56-6	Skin Irrit. 3 Aquatic Chronic 3 H316, H412
9,11-Octadecadienoic acid, (9Z,11E)- Content (W/W): $\geq 25\%$ - $< 50\%$ CAS Number: 2540-56-9	Skin Irrit. 3 Aquatic Chronic 3 H316, H412
Ethyl linoleate Content (W/W): $> 0\%$ - $< 3\%$ CAS Number: 544-35-4 EC-Number: 208-868-4	Skin Corr./Irrit. 2 H315

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:

Wash thoroughly with soap and water

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:

water spray, foam, carbon dioxide, dry powder

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture

Acrylaldehyde, carbon oxides, harmful vapours

The substances/groups of substances mentioned can be released in case of fire. Generation of fumes/fog.

Advice for fire-fighters

Special protective equipment:

Wear a self-contained breathing apparatus.

Further information:

Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Cool endangered containers with water-spray.

6. Accidental Release Measures

High risk of slipping due to leakage/spillage of product. Soiled textiles/cleaning rags made of natural fibres (e.g. of pure wool or of pure cotton) are capable of ignition and should not be used and/or must be disposed of in a safe manner.

Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Information regarding personal protective measures, see section 8.

Environmental precautions

Do not discharge into drains/surface waters/groundwater. Inform authorities in the event of product spillage to water courses or sewage systems.

Methods and material for containment and cleaning up

For small amounts: Pick up with suitable absorbent material. Do not use saw-dust or other combustible substances as an absorbant during cleanup.

For large amounts: Dike spillage. Pump off product.

Dispose of absorbed material in accordance with regulations. Mop up spills with non-flammable adsorbents (e.g. vermiculite, spill mats). Soiled textiles / cleaning rags / adsorbents and Silica are capable of self ignition and should be wetted with water and must be disposed of in a safe manner.

7. Handling and Storage

Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:

Risk of self-ignition when a large surface area is produced due to fine dispersion. Soiled textiles / cleaning rags / adsorbents and Silica are capable of self ignition and should be wetted with water and must be disposed of in a safe manner. Take precautionary measures against static discharges. Avoid all sources of ignition: heat, sparks, open flame.

Conditions for safe storage, including any incompatibilities

Suitable materials for containers: Stove-lacquer R 78433, High density polyethylene (HDPE), Paper/Fibreboard

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

57-11-4: Stearic acid

112-80-1: Oleic acid

Exposure controls

Personal protective equipment

Respiratory protection:

Respiratory protection in case of vapour/aerosol release. Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:

Chemical resistant protective gloves (EN ISO 374-1)

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:

Body protection must be chosen based on level of activity and exposure.

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is recommended. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Store work clothing separately.

9. Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter: liquid

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Form:	oil, liquid	
Colour:	colourless to slightly yellow	
Odour:	almost odourless	
Odour threshold:		
	not determined	
Melting point:	5 °C	
Boiling point:	412 °C (1.013,3 hPa)	
	The product has not been tested., Literature data.	
Flammability:	hardly combustible	
Lower explosion limit:		
	For liquids not relevant for classification and labelling.	
Upper explosion limit:		
	For liquids not relevant for classification and labelling.	
Flash point:	> 250 °C	(ISO 2719)
Auto-ignition temperature:		
	Study does not need to be conducted.	
Thermal decomposition:	>= 100 °C (DSC (DIN 51007))	
pH value:		
	insoluble	
Viscosity, kinematic:	approx. 74 mm ² /s (20 °C)	(calculated (from dynamic viscosity))
Viscosity, dynamic:	66 mPa.s (20 °C)	
Solubility in water:	insoluble	
Solubility (qualitative) solvent(s):	Ethanol partly soluble	
Solubility (qualitative) solvent(s):	organic solvents soluble	
Partitioning coefficient n-octanol/water (log K _{ow}):		
	not applicable for mixtures	
Vapour pressure:	< 1,333 hPa (20 °C)	
	The product has not been tested., Literature data.	
Density:	0,891 g/cm ³ (20 °C)	
Relative vapour density (air):	> 1 (20 °C)	(estimated)
	Heavier than air.	

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: Based on the chemical structure
there is no indication of explosive
properties.

Oxidizing properties

Fire promoting properties: not fire-propagating

Self-heating substances and mixtures

Self heating ability: not applicable, the product is a liquid

Corrosion to metals

Corrosive effects to metal are not anticipated.

Other safety characteristics

SAPT-Temperature:

Study scientifically not justified.

Evaporation rate:

not determined

10. Stability and Reactivity**Reactivity**

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: Corrosive effects to metal are not anticipated.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

When finely distributed, self-ignition is possible.

Conditions to avoid

See SDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:

oxidizing agents, bases, strong acids

Hazardous decomposition products

Hazardous decomposition products:

No hazardous decomposition products known.

11. Toxicological Information**Information on toxicological effects**Acute toxicity

Assessment of acute toxicity:

Virtually nontoxic after a single ingestion.

Irritation

Assessment of irritating effects:

Avoid contact with the skin, eyes and clothing.

Respiratory/Skin sensitization

Assessment of sensitization:

Based on available data, the classification criteria are not met.

Germ cell mutagenicity

Assessment of mutagenicity:

Based on available data, the classification criteria are not met. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:

Not classified, due to lack of data.

Reproductive toxicity

Assessment of reproduction toxicity:

Not classified, due to lack of data.

Developmental toxicity

Assessment of teratogenicity:

Not classified, due to lack of data.

Specific target organ toxicity (single exposure)

Remarks: Based on available data, the classification criteria are not met.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Not classified, due to lack of data.

Aspiration hazard

No data available.

Other relevant toxicity information

The product has not been tested. The statement has been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:

Harmful to aquatic organisms based on long-term (chronic) toxicity study data.

Information on: 9,11-Octadecadienoic acid, (9Z,11E)-

Toxicity to fish:

LC50 (48 h) > 10.000 mg/l, *Leuciscus idus* (OECD Guideline 203, static)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

LC50 (96 h) > 2,5 mg/l, *Oryzias latipes* (OECD Guideline 203)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: 10,12-Octadecadienoic acid, (10E,12Z)-

Toxicity to fish:

LC50 (48 h) > 10.000 mg/l, *Leuciscus idus* (OECD Guideline 203, static)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

LC50 (96 h) > 2,5 mg/l, *Oryzias latipes* (OECD Guideline 203)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: 9,11-Octadecadienoic acid, (9Z,11E)-

Aquatic invertebrates:

EC50 (47 h) > 32 mg/l, *Daphnia magna* (Directive 92/69/EEC, C.2, static)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

LC50 (48 h) > 20 mg/l, *Artemia salina* (other)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: 10,12-Octadecadienoic acid, (10E,12Z)-

Aquatic invertebrates:

EC50 (47 h) > 32 mg/l, *Daphnia magna* (Directive 92/69/EEC, C.2, static)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

LC50 (48 h) > 20 mg/l, *Artemia salina* (other)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: 9,11-Octadecadienoic acid, (9Z,11E)-

Aquatic plants:

EC50 (72 h) > 2,6 mg/l, *Pseudokirchneriella subcapitata* (OECD Guideline 201)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

No observed effect concentration (72 h) 2,6 mg/l, *Pseudokirchneriella subcapitata* (OECD Guideline 201)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

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Information on: 10,12-Octadecadienoic acid, (10E,12Z)-

Aquatic plants:

EC50 (72 h) > 2,6 mg/l, *Pseudokirchneriella subcapitata* (OECD Guideline 201)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

No observed effect concentration (72 h) 2,6 mg/l, *Pseudokirchneriella subcapitata* (OECD Guideline 201)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: 9,11-Octadecadienoic acid, (9Z,11E)-

Microorganisms/Effect on activated sludge:

EC10 (18 h) 883 mg/l, *Pseudomonas putida* (DIN EN ISO 10712)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: 10,12-Octadecadienoic acid, (10E,12Z)-

Microorganisms/Effect on activated sludge:

EC10 (18 h) 883 mg/l, *Pseudomonas putida* (DIN EN ISO 10712)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: 9,11-Octadecadienoic acid, (9Z,11E)-

Chronic toxicity to aquatic invertebrates:

No observed effect concentration (21 d) 0,32 mg/l, *Daphnia magna* (OECD Guideline 211)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: 10,12-Octadecadienoic acid, (10E,12Z)-

Chronic toxicity to aquatic invertebrates:

No observed effect concentration (21 d) 0,32 mg/l, *Daphnia magna* (OECD Guideline 211)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Assessment of terrestrial toxicity:

No data available.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):

Biodegradable.

Bioaccumulative potential

Assessment bioaccumulation potential:

The product has not been tested.

Mobility in soil

Assessment transport between environmental compartments:

Volatility: No data available.

Adsorption in soil: No data available.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria. Self classification

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:

The product has not been tested. The statement has been derived from the properties of the individual components. Do not allow to enter soil, waterways or waste water channels.

13. Disposal Considerations

Waste treatment methods

Observe national and local legal requirements.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

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	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information**Safety, health and environmental regulations/legislation specific for the substance or mixture**

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

Any other intended applications should be discussed with the manufacturer. Corresponding occupational protection measurements must be followed.

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Aquatic Chronic	Hazardous to the aquatic environment - chronic
Skin Irrit.	Skin irritation
Skin Corr./Irrit.	Skin corrosion/irritation
H316	Causes mild skin irritation.
H412	Harmful to aquatic life with long lasting effects.
H315	Causes skin irritation.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.